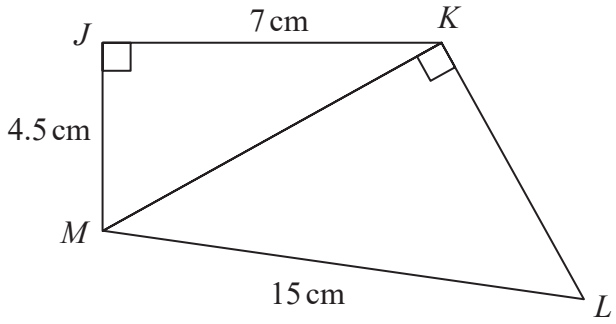
6 The diagram shows a quadrilateral *JKLM*.



Work out the size of angle *KLM*.
Give your answer correct to 3 significant figures.

Triangle JKM is right angled at J.

$KM^2 = 7^2 + 4.5^2 = 69.25$.

1 mark
Start Pythagoras

$KM = \sqrt{69.25} = 8.321658489...$cm.

1 mark
Find KM

Triangle KLM is right angled at K; ML=15cm.

$\sin(\text{angle KLM}) = KM \div ML = \sqrt{69.25} \div 15$.

1 mark
Opposite ÷ hypotenuse

$\text{Angle KLM} = \sin^{-1}(\sqrt{69.25} \div 15) = 33.696...°$.

33.7° (3 significant figures).

1 mark
33.7°

°

(Total for Question 6 is 4 marks)